

Indian Institute of Information Technology Bhopal

Mid-Semester Examination, March 2026

Discrete Structure

CSE/CSE(AI)/CSE(CS)/CSE(CPS)/CSE(DS)-2002

Semester: II

Time: 90 Min

Max. Marks: 20

Instructions:

- All questions are compulsory.
- Mark answers number clearly.

1. (a) Use strong mathematical induction to prove that every integer $n \geq 12$ can be written in the form

$$n = 4a + 5b,$$

where a and b are non-negative integers.

- (b) Let $A = \{1, 2, 3, 4, 5\}$, $B = \{a, b, c\}$. Using the Principle of Inclusion-Exclusion, determine the number of functions $f : A \rightarrow B$ such that

$$f(1) \neq a, \quad f(2) \neq b, \quad f(3) \neq c.$$

[2+3]

2. (a) State whether the following statements are true or false. Give brief justification.

- Every finite lattice has a least element and a greatest element.
- Every bounded lattice is necessarily finite.

- (b) Let $S = \{(a, b) \in \mathbb{Z} \times \mathbb{Z} \mid (a, b) \neq (0, 0)\}$. Define a relation $\sim$ on S by

$$(a, b) \sim (c, d) \quad \text{if and only if} \quad ad - bc = 0.$$

- Prove that $\sim$ is an equivalence relation on S .

- ii. Describe the equivalence class of an arbitrary element $(a, b) \in S$.

[2+3]

3. Let $X = \{1, 2, 3, 4, 5\}$ and consider the power set $\mathcal{P}(X)$ ordered by set inclusion $\subseteq$.

- (a) Prove that $(\mathcal{P}(X), \subseteq)$ is a partially ordered set.
- (b) Determine all minimal elements and maximal elements of this poset.
- (c) Identify the least element and the greatest element.
- (d) Let $A = \{1, 2\}$ and $B = \{2, 3, 4\}$. Find all upper bounds and all lower bounds of $\{A, B\}$.
- (e) Determine the least upper bound (LUB) and the greatest lower bound (GLB) of $\{A, B\}$.

[5]

4. (a) Let $\mathbb{Z}_n = \{0, 1, 2, \dots, n-1\}$ be the set of integers modulo n , with addition and multiplication defined modulo n .

- i. Determine whether $\mathbb{Z}_{12}$ is an integral domain.
- ii. Determine all values of n for which $\mathbb{Z}_n$ forms a field. Justify your answer.

- (b) Let $U(15) = \{1 \leq x \leq 15 \mid \gcd(x, 15) = 1\}$ be the group of units modulo 15 under multiplication.

- i. Construct the multiplication table of $U(15)$.
- ii. Find the inverse of each element in $U(15)$.
- iii. Determine whether $U(15)$ is cyclic. Justify your answer.

[2+3]